

OAGRE: Outlier Attenuated Gradient Boosted Regression

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Abstract Gradient boosting is a general machine learning technique for iteratively building a model by fitting the residuals of an existing model. In spite of its generality as a technique, there are several reasons why gradient boosting may fail to adequately model a given dataset. One of these reasons is the presence of excessive noise in the training data. In this paper we propose a technique that will mitigate the impact of noise under some conditions of heteroscedasticity. We break the assumption of independently distributed noise and derive a simple variation of the gradient boosting algorithm we name Outlier Attenuated Gradient Boosting Regression (OAGRE). We conduct experiments by generating large volumes of synthetic datasets with varying degrees of heteroscedasticity and benchmark our approach against multiple standard gradient boosting algorithms as well as simpler tree models. Our approach provides value in approx 30% of these synthetic datasets with an average improvement of between 7% and 12% lower root mean squared error. Based on these results we recommend the inclusion of the OAGRE method when testing algorithms on data with suspected heteroscedasticity issues.

1 Introduction

Gradient Boosting has proven itself to be one of the most significant advances in the field of machine learning in recent history. Its utility in providing accurate predictions for supervised learning is so strong that it has come to dominate the field of competitive machine learning on tabular datasets (as exemplified by the competitions run by Kaggle [1]). In spite of its success, there remain a range of specific circumstances which limit the ability of gradient boosting (and other algorithms) to provide adequate results. One of these circumstances is the presence of complex noise in the

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data. Early boosting algorithms, like AdaBoost, were known to suffer in the presence of noise [2, 3]. Modern gradient boosting algorithms are less susceptible to these issues and tend to perform well with small amounts of heteroscedastic noise [4]. However, there is always potential for learning issues to emerge if some of the residuals become dominated by the presence of noise.

Gradient boosting algorithms are predominantly tree based and there are multiple variations offering improvements in performance and computational efficiency, typically through the way variables are selected and the base learners are combined [5]. However, in all cases the theoretical underpinnings of the gradient boosting algorithm remain unchanged. The algorithm consists of learning an ensemble of base learners as an additive combination. In each iteration the new model is trained to predict the residuals of the preceding ensemble [6, 7, 8].

In this paper we develop a variation of the gradient boosting algorithm by making several non-standard assumptions about the noise present in the data. This involves breaking the assumption that the noise is identically distributed across the problem space. In particular we assume that the distribution of the noise varies in a way that can be at least partially predicted by the known regressors for the machine learning problem.

Noisy data with heteroscedasticity has been shown to be deleterious to regression trees (the most common base learner for gradient boosting models) [9]. In addition, there are many application domains in which the noise is known to be heteroscedastic, and the focus of research is to develop custom algorithms that explicitly model the noise variance [10, 11], or build custom ensembles where subcomponents are used to accommodate the heteroscedasticity [12]. Common general machine learning approaches include the development of noise filtering or re-labelling techniques, under the assumption that the noisy data is an outlier to be detected [13]. Other research has focused on modification of the base learners in ensemble methods (including gradient boosting ensembles) to make them more robust to heteroscedasticity [14], or to attempt to explicitly model the heteroscedastic noise distribution [15, 16] and incorporate the noise distribution into the loss function to weight samples accordingly [17]. An alternative approach has been to augment the design of the gradient boosting method by quantifying the uncertainty in the predictions generated by the heteroscedasticity [4], or to model the output distribution directly using quantile or expectile regression [18]. In situations in which there is known noiseless data, or it can be simulated, then there are techniques that allow models to be trained such that they are influenced less by the noise distribution [19]. However, as our ability to fit models depends heavily on the amount of data and the nature of the noise, which is usually unknown, heteroscedasticity remains an open problem for machine learning research.

In this work we explore the extent to which assuming heteroscedastic noise in a dataset can be used to derive a fundamental variation of the gradient boosting algorithm. The core idea is that if the noise distribution is variable then high noise regions of the problem space should be fit with simpler models than the low noise regions. Such that the model does not waste computational resources fitting noisy data points. The gradient boosting is selectively applied by predicting whether a

given sample belongs to a region that surpasses the noise threshold for each level of the boosted model.

To do so we assume that there will be some correlation between the magnitude of the error made by the model and the presence of higher variance noise. This assumption requires only that our feature vector and model be rich enough that we expect the model to be able to learn the relationship between dependent and independent variables well enough that noisier samples stand out. In the absence of noise we expect error to be low, when noise is present we expect it to be high. Because the first assumption gives us non-uniform noise, we therefore expect the presence of a noise gradient that is correlated with the error.

Combining these two assumptions with the learning strategy of gradient boosting for regression, leads us to an alternative formulation of the gradient boosting algorithm. The core idea being that each iteration of boosting should seek to learn only from residuals that come from regions in the parameter space that contain less noise. In other words we predict which samples are noisier outliers and exclude them from the training of subsequent models in the ensemble. We call this algorithm Outlier Attenuated Gradient Boosting Regression (OAGRE)

2 Method

We start the derivation of the method by assuming that the noise is not randomly distributed with respect to the features or predictors, but instead is predictably heteroscedastic. This assumption is contrary to the standard expression for the learning problem, an example of which is shown in Equation 1 taken from Friedman 2000 [6].

$$y_i = F^*(X_i) + \varepsilon_i \quad (1)$$

This expression defines the target variable of instance i , y_i as a function of the features vector X plus some Gaussian noise ε_i . It is generally assumed that the noise term is independent of the feature vector X . It may correspond to inherent noise, or be due to unobserved causal factors that are not present in the features, and hence limit our ability to perfectly represent the target using the given features.

We modify this assumption in the following way: the noise term ε_i is partially dependent upon the feature vector X_i , but only in its variance. In real world processes there are multiple reasons why noise might be correlated with predictors. Noise can come from sampling problems, which are not uniformly distributed. Noise might be due to limitations of the devices or methodology used to collect data, which again tend not to be uniformly distributed. Most noise is due to some physical process, which itself is regulated by variables, if any of the variables that regulate the noise are present as features then we should expect the statistical structure of the noise to vary over the features space.

This means that we should expect the learning problem in many real world problems to be

$$y_i = F^*(X_i) + \varepsilon(N(X_i)) \quad (2)$$

in which $\varepsilon(N(X_i))$ refers to a sample drawn from a noise distribution N that is dependent on the independent variables of the sample X_i .

When we are training a model to predict y_i , then we typically depict the fit of the model using a similar equation.

$$\hat{y}_i = \hat{F}(X_i) \quad (3)$$

Such that the function $\hat{F}$ has been fit such that given a feature vector X_i it provides an estimate of y_i . The error of such a model can be calculated for each record as the difference between the actual and predicted value:

$$err_i = y_i - \hat{y}_i \quad (4)$$

Note that error is typically calculated using some loss function $L(y, \hat{y})$, however we will ignore that complication for the moment. We can substitute Equations 2 and 3 into Equation 4 and with some rearrangement end up with the following expression for the error:

$$err_i = (F^*(X_i) - \hat{F}(X_i)) + \varepsilon(N(X_i)) \quad (5)$$

In other words the expected error in our model is the difference between the true underlying function and the estimated one, plus the expected noise in that region of the problem space.

For many algorithms the change we have made to the distribution of noise is not something that can easily be exploited. However, the gradient boosting algorithm has the potential to exploit this new feature of the problem. Usually noise undermines a gradient boosting step because it is attempting to use the residual of the existing model to build a new iteration that improves the previous model. One might imagine that at some point the residuals come to be dominated by the noise term, and hence a gradient boosted algorithm cannot see the signal for the noise.

If however, the noise is not identically distributed across the feature space then we could potentially direct each round of gradient boosting to ignore noisy examples, and focus on learning from a subset of the data. This is similar to what Freund did in the development of BrownBoost [2], a generalisation of AdaBoost, although in his treatment it involved setting an expected minimum error rate and then smoothly downweighting samples that exceeded this threshold.

In order to modify the learning algorithm to ignore samples suspected of being dominated by noise we need to provide some kind of selection criteria for deciding at the start of each iteration which samples to exclude. We introduce an additional threshold meta-parameter which determines when an example should be ignored. This parameter determines the threshold in the standard deviation of the residual errors beyond which a sample is considered an outlier. In addition we allow this parameter to be adjusted downward with each iteration of learning, so that the model

increases the volume of samples that are treated as outliers as the boosting ensemble grows.

In order to apply the model to new data we cannot simply make predictions using all stages of the gradient boosting because deeper level trees have been trained only on samples from a subset of the feature space. In other words, our gradient boosted model covers the problem space at different levels of granularity, depending on our expectations of the noise distribution. We only apply each subsequent gradient boosting model in a way that it is attenuated according to our estimate of whether or not it is appropriate. We do this by fitting a classification model at each stage of gradient boosting which predicts whether or not a given example belongs to the signal set (of relatively low noise examples). We use the classifier to build a model to estimate the probability that sample i is considered to be low noise at depth j in our gradient boosted model: $p_j(\text{lownoise}|X_i)$.

The final prediction can be calculated according to the following formula:

$$\hat{y}_i = \hat{B}(X_i) - \sum_{j=1}^M p_j(\text{lownoise}|X_i) * \hat{R}_j(X_i) \quad (6)$$

where M is the parameter defining the number of models in the ensemble, $\hat{B}$ is the base regression model fitted to the entire dataset, and $\hat{R}_j$ is the estimate of the residuals at depth j of the boosting. Our prediction $\hat{y}_i$ is a combination of the base model with the negative sum of all residual estimates over the M levels of boosting each of which is attenuated by the estimate of the probability that the sample is a low noise example at that depth of the boosted model.

2.1 Implementation

```

1 from sklearn.tree import DecisionTreeClassifier
2 from sklearn.tree import DecisionTreeRegressor
3 from oagre import OAGRE
4
5 oagre = OAGRE(
6     classifier=DecisionTreeClassifier(max_depth=5),
7     regressor=DecisionTreeRegressor(max_depth=5)
8 )
9
10 oagre.fit(X_train, y_train)
11 ...

```

Listing 1 Usage of OAGRE with scikit-learn models

We have implemented the OAGRE method as a meta-learning algorithm compatible with scikit-learn [20], and made it available as an open source python library in the GitHub repository (<https://github.com/john-hawkins/oagre>). The implementation requires that the user provide two base models, a classifier and a regressor, which it then uses to building the gradient boosted ensemble. We show an example

of how to instantiate and train an OAGRE regressor in Listing 1, where we use the sklearn DecisionTree models for the base classifier and regressor.

Our implementation allows an OAGRE ensemble model to be included inside the machine learning framework scikit-learn pipelines [20]. The advantage of the pipeline approach is that an OAGRE model can be built into a single serialised model for deployment, including data preprocessing stages. The OAGRE library can be installed as a package from the PyPi repository (<https://pypi.org/project/oagre>) or installed directly from the source code (<https://github.com/john-hawkins/oagre>).

2.2 Evaluation

To evaluate the effectiveness of our approach we conduct an experiment in which large volumes of synthetic data sets are created with varying parameterisation of the heteroscedasticity in the error term. The parameters of the simulation are shown in Table 1. We generate 100 random datasets for each combination of parameters, such that the independent variables are drawn from a range of different numerical distributions and the target variable is created through a randomly generated formula combining linear and non-linear combination of a random subset of independent variables. We generate a second random function that is fed through a logistic function to create a probability distribution. The outputs of this probability distribution is used to determine which datapoints will be subject to excessive noise, thus ensuring that the data is not only heteroscedastic, but that this property is partially dependent on the independent variables.

Once the data is generated we take a random split of 90% for training and 10% for testing. We train six models for evaluation and comparison. The first two are standard implementations of the Gradient Boosted Tree algorithm, the GBM model built into the scikit-learn package, and the LightGBM model [5]. For each of these models we train them use default parameters, in order to reduce the computational complexity of performing hyperparameter tuning over large volumes of synthetic data. The second two models are OAGRE ensembles using different base learners. The first is trained with an ExtraTrees model (a Random Forest Implementation) and the second with a DecisionTree model, both of these are taken from the standard scikit learn implementation. Finally, we included the base regression models used by the two OAGRE models for comparison, to ensure that the OAGRE algorithm was adding value to these models. The standard GBM models were initialised with default parameters, the other models were initialised shown in Table 2. These initialisations are based on our experience with applying the OAGRE technique. Note, that when using the ExtraTrees model as a base learner, the number of estimators is a combination of the estimators used by ExtraTrees and the estimators used in the boosting layers of OAGRE.

The complete code for the simulation and evaluation is provided in our GitHub repository. We note that as the generation of random formulas sometimes produced null values, there is significant checks in place to regenerate data when problematic

Table 1 Simulation Parameters

Parameter	Description	Values
Dataset Size	Number of records in the dataset	2000, 5000, 10000
Variables	Number of independent variables	15, 20, 25, 30
Outlier Proportion	Proportion of records with larger noise	0.4, 0.5, 0.6, 0.7
Std Multiplier	The factor applied to noise std of outliers	random int(3,8)

scenarios occur. In total this process produced 2,185 datasets that the models were trained and evaluated on.

Table 2 Models and Initialisation Parameters

Model	Description	Estimators	Max Depth
GBM	SKLearn GB Regressor	100	3
LGBM	LightGBM	100	Unbounded
OARGE-DT	OAGRE with DecisionTrees	20	5
OAGRE-XT	OAGRE with ExtraTrees	200	5
XTR	ExtraTrees Regression	10	5
DTR	DecisionTree Regression	1	5

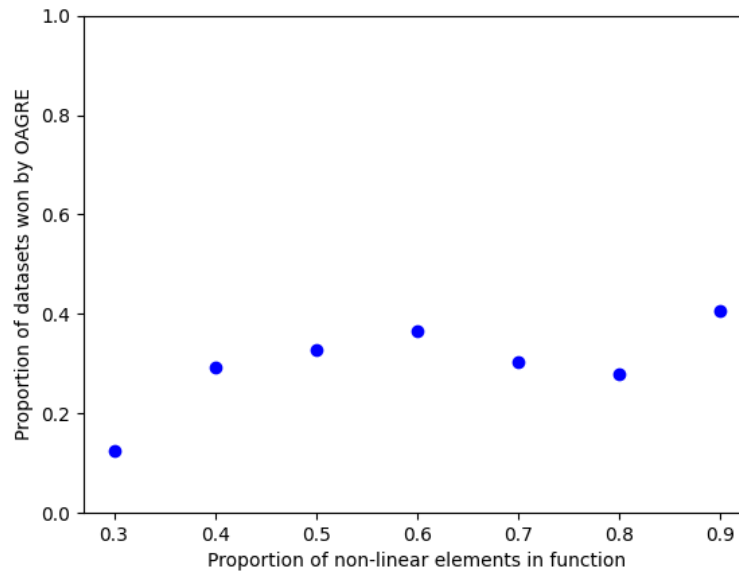
3 Results

In Table 3 show the proportion of datasets for which each model provided the best result (measured in terms of lowest Root Mean Squared Error). In addition, we show the average percentage improvement of each model against the next best model for the task. As we see, the overall winner is the standard Gradient Boosting Machine implementation from the scikit-learn library. The two OAGRE implementations provide a superior model in approx 30% of datasets, with the version using ExtraTrees models as the base learner showing stronger performance, both in terms of the proportion of times it is superior, and the expected percentage of improvement. The two base models alone are only superior in a small percentage of instances, with a more marginal improvement.

We took the parameters that generated these synthetic datasets and analysed multiple factors that might contribute to the circumstances in which the OAGRE approach added value. This included the outlier proportion and standard deviation multiplier factor used to generate the data, as well as the proportion of non-linear elements used in the random formula that generated the target variable. We observed

Table 3 Overall Results

Model	Total Wins	Percentage of Wins	Average Improvement
GBM	890	40.7%	30.8%
LGBM	274	12.5%	25.4%
OARGE-DT	279	12.7%	11.6%
OAGRE-XT	430	19.7%	7.3%
XTR	131	6.0%	2.5%
DTR	180	8.2%	2.4%

**Fig. 1** OAGRE Wins against Non-linear Elements in Target Function

that the proportion of datasets upon which OAGRE provided superior performance appeared to be increasing with the proportion of the generating function that was non-linear, as shown in Figure 1.

4 Conclusion

By questioning a standard assumption in the definition of regression problems, pertaining to the nature of the noise present in a data set, we have been able to derive a

variation of the gradient boosting algorithm. Our variation compensates for the presences of heteroscedasticity by explicitly modelling the probability that a data point comes from a high noise region of the data space, and then excluding it from the residual estimates in the boosting process. We incorporate the probability of a data point being a noisy outlier at prediction time to attenuate the impact of residual estimates in the boosting process. The resulting meta-learning approach to building an ensemble we call Outlier Attenuated Gradient Boosting Regression and we release it as a scikit-learn compatible open source library.

We conducted a simulation based experiment in which we generated large volumes of datasets with heteroscedastic noise imposed through parameterisation. These experiments indicate that our approach will provide superior performance to standard gradient boosting implementation in approx 30% of cases, with an average reduction in root mean squared error of between 7% and 12% (compared to the next best model). We note that this work was done using default parameters to limit the complexity of the search space, hence the impact of significant parameter tuning is unknown, but would likely to shift some of these results. Nevertheless, we observe that the proportion of instances in which the OAGRE approach proved superior was approximately equal to its proportion in the set of methods tested. Hence, we conclude that the technique will likely provide value for some regression problems with heteroscedastic noise. For this reason we recommend that practitioners facing a regression problem with noisy heteroscedastic data include the OAGRE algorithm within a set of empirically evaluated methods.

Future research could be conducted to determine the exact properties of a dataset that make it amenable to our technique. However, preliminary analysis suggests that the more non-linear the function that is being estimated, the greater the likelihood that OAGRE will provide predictive value.

References

1. A. Fogg, “Anthony Goldbloom gives you the secret to winning Kaggle competitions.” Blog-post: <https://www.import.io/post/how-to-win-a-kaggle-competition/>, 2016.
2. Y. Freund, “An adaptive version of the boost by majority algorithm,” *Machine Learning*, vol. 43, no. 3, pp. 293–318, 2001.
3. R. E. Schapire, “The boosting approach to machine learning: An overview,” in *Nonlinear Estimation and Classification*, pp. 149–171, New York, NY: Springer New York, 2003.
4. J. Brophy and D. Lowd, “Instance-based uncertainty estimation for gradient-boosted regression trees,” in *Proceedings of the 36th Conference on Neural Information Processing Systems (NeurIPS 2022)*, 2022.
5. G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS’17*, (Red Hook, NY, USA), p. 3149–3157, Curran Associates Inc., 2017.
6. J. Friedman, T. Hastie, and R. Tibshirani, “Additive logistic regression: A statistical view of boosting,” *The Annals of Statistics*, vol. 28, pp. 337–407, 2000.
7. J. Friedman, “Greedy function approximation: A gradient boosting machine,” *The Annals of Statistics*, vol. 29, no. 5, p. 1189–1232, 2001.

8. J. Friedman, "Stochastic gradient boosting," *Computational Statistics & Data Analysis*, vol. 38, pp. 367–378, 2002.
9. W. Ruth and T. Loughin, "The effect of heteroscedasticity on regression trees," 2016.
10. L. Zhenxing, H. Xiaodan, K. Hao, L. Chen, and B. Huang, "Gaussian process regression with heteroscedastic noises – a machine-learning predictive variance approach," *Chemical Engineering Research and Design*, vol. 157, 2020.
11. J. Zhang, F. Guo, K. Hao, B. Huang, and L. Chen, "Identification of errors-in-variable system with heteroscedastic noise and partially known input using variational bayesian," *IEEE Transactions on Industrial Informatics*, vol. 19, no. 10, pp. 10014–10023, 2023.
12. H. Zeng, B. Shao, G. Bian, H. Dai, and F. Zhou, "A hybrid deep learning approach by integrating extreme gradient boosting-long short-term memory with generalized autoregressive conditional heteroscedasticity family models for natural gas load volatility prediction," *Energy Science & Engineering*, vol. 10, 2022.
13. Y. Ustinovskiy, V. Fedorova, G. Gusev, and P. Serdyukov, "Meta-gradient boosted decision tree model for weight and target learning," in *Proceedings of the 33rd International Conference on International Conference on Machine Learning - Volume 48*, ICML'16, p. 2692–2701, JMLR.org, 2016.
14. A. J. Henrey, *Statistical Learning Tools for Heteroskedastic Data*. Phd thesis, Simon Fraser University, 2016.
15. R. Natarajan, "Gradient boosting for joint regression modeling of mean and dispersion," Research Report W0909-085, IBM Research Division, New York, 2009.
16. J. Thomas, A. Mayr, B. Bischl, M. Schmid, A. Smith, and B. Hofner, "Gradient boosting for distributional regression: faster tuning and improved variable selection via noncyclical updates," *Statistics and Computing*, vol. 28, pp. 1–15, 2018.
17. G.-H. Lee, S.-W. Yang, T.-H. Lin, and S.-D. Lin, "Heteroscedastic learning and its realization on matrix factorization and gradient boosting regression/classification," tech. rep., 2017.
18. Y. Yang and H. Zou, "Nonparametric multiple expectile regression via er-boost," *Journal of Statistical Computation and Simulation*, vol. 85, pp. 1–17, 2014.
19. Z. Wu, J. Luo, D. Rincon, and P. Christofides, "Machine learning-based predictive control using noisy data: Evaluating performance and robustness via a large-scale process simulator," *Chemical Engineering Research and Design*, vol. 168, 2021.
20. F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011.